

NTT Procurement Gameplan

NTT Procurement Intelligence Workstream

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Scope and Approach

Context and Problem Statement

NTT DATA Procurement operates at global scale, managing approximately \$11.0B in spend under management, ~250,000 purchase orders per year, ~1.9 million invoice lines, operations across 54 countries, and a procurement organization of roughly 470 professionals. In this document, the focus is the procurement workstream and the Horizon 1 foundations required to establish an intelligence-led operating model for the function.

Within NTT DATA, procurement currently operates across a fragmented estate of 40+ source systems, including multiple SAP instances and two Ariba environments. The function contains strong operational expertise, but the current model is constrained by incomplete classification, duplicate supplier entities, contracts trapped in shared drives, and limited cross-ERP visibility. As a result, answering basic questions on spend, supplier exposure, contract coverage, and route-to-action still requires significant manual reconciliation.

Horizon 1 should be anchored on three data foundations and one visible proof workflow: **Spend Lake, Supplier Master + ESG, Contract Intelligence**, and the User Experience plane ie. **The Front Door**. These are not isolated use cases; they form the minimum cross-ERP data layer and routing capability required to prove the broader intelligence-led procurement architecture.

NTT DATA also has an explicit vision for **Intelligence-Led Procurement**: an AI intelligence layer that sits above systems of record, unifies fragmented procurement data, and improves how the function sees, structures, and routes work. For Horizon 1, the procurement workstream should establish a practical operating surface that combines grounded data foundations, workflow orchestration, and human-in-the-loop AI support rather than treating AI as a generic assistant layer.

The problem, therefore, is not simply that procurement processes are manual. It is that procurement lacks a reliable, cross-ERP intelligence layer that can classify spend at scale, resolve supplier entities globally, turn contracts into queryable assets, and route incoming demand using evidence rather than fragmented local knowledge. Without that foundation, category strategy, supplier oversight, contract leverage, and requester guidance remain slower, less consistent, and more reactive than they should be.

1.1.1. Procurement Horizon 1 Lifecycle

A Horizon 1 procurement journey starts with a **demand signal** from the business: a request for goods, services, labor, or supplier engagement. In the current model, that signal is often shaped differently by region, system, and business unit before it reaches procurement, making early classification and policy-aware routing inconsistent.

The first requirement is **spend visibility and classification**: ingesting spend records from multiple ERP systems, normalizing them, and classifying them to the NTT DATA taxonomy with confidence scoring. This is the foundation that makes category insight, benchmarking, anomaly detection, and cross-ERP questioning possible.

Supplier resolution and enrichment then becomes critical. The same supplier may exist multiple times across ERPs under different identifiers, with incomplete parent-child relationships and fragmented risk context. Horizon 1 addresses this by creating one global supplier view for the top supplier population and enriching it with ESG, risk, and performance signals.

The third requirement is **contract intelligence**: bringing active supplier contracts out of shared drives, extracting key terms, indexing them for retrieval, and linking them to spend and supplier records. This creates the minimum viable contract layer needed for expiry monitoring, clause retrieval, and route selection.

Once those foundations exist, procurement can expose a **Front Door** that captures and structures incoming requests, consults spend, supplier, and contract intelligence in parallel, and routes each request to the most appropriate path: existing contract, catalog, RFx, or human assist.

Across this lifecycle, the goal is not just better reporting. It is to create a working intelligence layer that procurement users can query, trust, and use to act faster. Horizon 1 is therefore a foundation build with a workflow proof on top, not a point solution or dashboard project.

The core problem: procurement decisions are being made across fragmented systems without a shared intelligence layer for classification, supplier resolution, contract visibility, and guided routing. Horizon 1 is designed to address that gap by giving procurement a cross-ERP data foundation and a visible workflow that proves it works.

Current Pain PointsPlease add optimization@Kalyan Chakravadhanula , added 8)

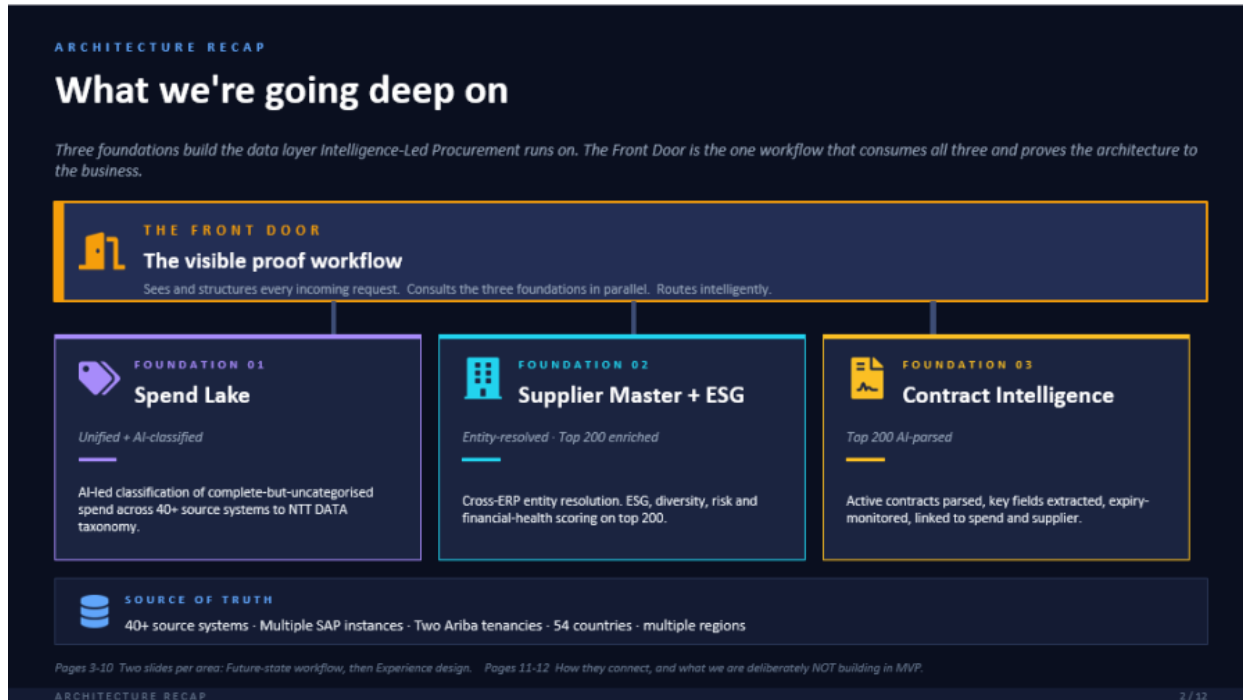
The workshop and draft backlog point to a specific set of procurement pain points that should anchor this document, rather than a generic transformation narrative.

1. Lack of cross-ERP spend visibility and taxonomy classification. Spend exists, but category-level understanding is incomplete because the data is not consistently classified.
2. Supplier entities are fragmented across systems. Duplicate supplier records, missing parent-child relationships, and incomplete enrichment weaken supplier oversight and risk management.
3. Contracts are present but not operationally queryable. Key terms, expiry windows, obligations, and coverage remain locked inside documents and shared drives.
4. Incoming demand is not consistently structured at the point of entry. Requests arrive through multiple channels and require manual interpretation before routing can begin.
5. Cross-system orchestration is insufficient. Procurement teams rely too heavily on manual reconciliation and local knowledge to connect spend, supplier, contract, and request context.
6. The MVP question can easily drift toward isolated workflow optimization. The reference materials explicitly reposition the MVP as a foundation build with one visible proof workflow.
7. Role design and human-in-the-loop operations need to be designed in from the start, especially for review queues, confidence thresholds, and future operating roles.

8. Cost optimization signals are not surfaced consistently across personas. Global category leads lack a cross-region view of committed spend, duplicate vendors, and co-term opportunities needed for consolidation and negotiation leverage; regional procurement leads cannot easily detect off-contract pricing, price variance, or rolling-window savings opportunities; and business requesters do not have self-serve visibility into preferred suppliers, existing contract coverage, or realistic benchmark pricing before raising demand.

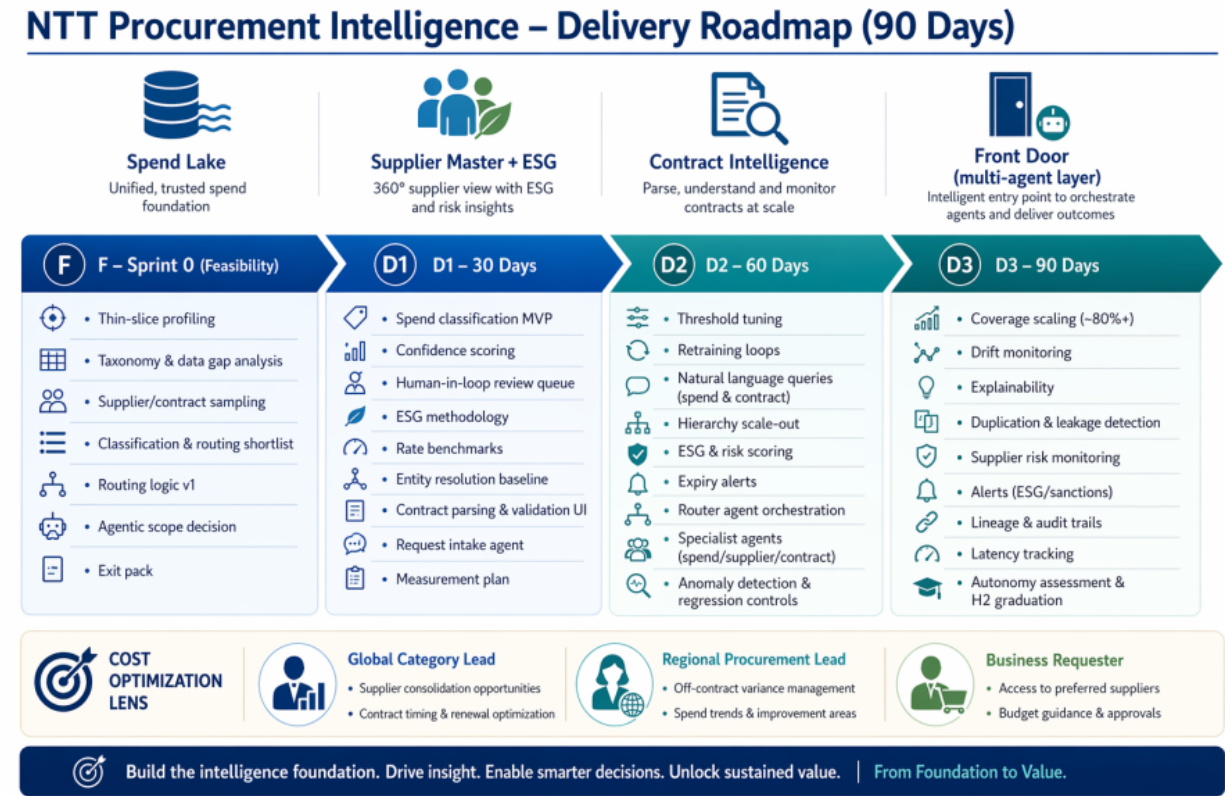
Objectives

The goal is to establish a procurement intelligence workstream that enables NTT DATA to unify spend, supplier, and contract intelligence across systems; route incoming demand with greater speed and consistency; and prove the value of an intelligence-led procurement operating model through a visible Horizon 1 workflow, while keeping humans firmly in control.



Roadmap Please add cost optimization in 90day bucket @Kalyan Chakravadhanula , please check the last para under 1.4.1. Also cost optimization user stories have been added(D3-US3 under D2-F4.5 · Contract Agent and D2-F1.3 · Natural Language Spend Query)

Roadmap phases are aligned to the uploaded delivery plan: **F** is the Feasibility Sprint 0, followed by **D1** (30 days), **D2** (60 days), and **D3** (90 days). The intent is to establish the three procurement foundations and the Front Door in a controlled sequence, starting with evidence-based feasibility, then progressing through production thin-slice delivery, scale-out, and operational hardening.



1.4.1. Delivery Roadmap

From feasibility evidence to an operational procurement intelligence foundation in 90 days

The roadmap should be read as a phased build of the Horizon 1 procurement intelligence layer. F serves as Sprint 0 and establishes the evidence base: thin-slice profiling, taxonomy and data-gap analysis, contract and supplier sample evaluation, classification and routing approach shortlists, routing logic v1, and the initial scope-in or scope-out decision for agentic patterns. This phase is intended to prove feasibility, surface material risks, and produce the exit pack that gates entry into delivery.

D1 (30 days) then moves the program into production thin-slice execution. The emphasis is on building the first usable spend-classification path, calibrating thresholds, standing up entity-resolution logic, defining ESG scoring methodology, producing initial rate benchmarks, extending profiling beyond the thin slice, validating extractor and routing quality, and attaching measurement plans to the scoped-in agentic patterns.

D2 (60 days) is the scale and convergence phase. Here the workstream should strengthen model and feature choices, complete leakage and regression controls, scale supplier hierarchy resolution, add supplier risk and performance scoring, operationalize retraining and anomaly detection, introduce fusion logic and failure handling for the Front Door, and formalize the autonomy ladder and readiness assessment for the agentic patterns that remain in scope.

D3 (90 days) focuses on operational hardening and outcome realization. By this stage, the workstream should push classification coverage toward the Day-90 target, introduce drift and model-health monitoring, complete hierarchy quality reporting, expose explainability and trend signals for supplier scoring, add duplicate and off-contract analytics, track end-to-end Front Door latency, and define graduation criteria and review cadence for H2+ agentic expansion.

Capabilities in this wave should support trend baselining, anomaly detection, stronger routing evaluation, readiness assessment for agentic patterns, and broader operationalization of model monitoring and explainability. Procurement leaders remain accountable for decisions, but the platform should increasingly surface risk, package evidence, and improve the speed and quality of intervention. PFB the planned features:-

A cost-optimization lens should cut across this roadmap from D1 onward. For the **Global Category Lead**, the priority is to surface committed spend, vendor overlap, and cross-region contract timing to enable consolidation and stronger negotiations. For the **Regional Procurement Lead**, the focus is on off-contract price variance, rolling-window spend views, and tactical self-serve answers that reduce manual analysis effort. For the **Business Requester**, the objective is to steer demand toward preferred suppliers, covered contracts, and realistic budgets before avoidable spend is committed.

Foundation 01 — Spend Lake (persona anchor: Hooman asks for historic plus committed spend by vendor/category/region; Stacey and business requesters ask for rolling-window top OEM/vendor views and off-contract checks)

D1-F1.1 · AI Spend Classification

- **D1-US1:** As a **Global Category Lead**, I want PO and invoice lines classified to the NTT DATA taxonomy by vendor, category, region, and time period so that I can answer "spend with this vendor in the last 12/18 months for this category and region" without manual report stitching.
- **D1-US2:** As a **Regional Procurement Lead**, I want "NTT Global" and other generic service descriptions decomposed into meaningful taxonomy labels so that recurring tactical questions are not trapped in unclassified spend buckets.
- **D1-US3:** As a **Business Requester**, I want plain-English spend answers such as "top 5/10 OEMs in the last 18 months" so that I can self-serve repeat questions without waiting for procurement.
- **D2-US1:** As a **Regional Procurement Lead**, I want rolling-window spend views across PO and invoice data, not only financial-year reports, so that leadership questions can be answered in the time frame they actually ask for.
- **D2-US2:** As a **Global Category Lead**, I want historic spend, committed PO value, and posted invoice spend shown side by side so that category planning is based on what has been spent and what has already been committed.
- **D3-US1:** As an **Auditor / Procurement Lead**, I want classification lineage, source attribution, model version, confidence, and override history on every transaction so that spend answers are explainable and traceable.

D1-F1.2 · Classification Feedback Loop (Human-in-the-Loop)

- **D2-US1:** As a **data scientist**, I want every analyst correction stored as a labelled training example so that weekly retraining uses real-world feedback.
- **D3-US1:** As a **data scientist**, I want accuracy tracked on a held-out audit sample so that I can measure model drift and improvement over time.

D1/D2-F1.3 · Persona-led Natural Language Spend Query

- **D1-US1:** As a **Regional Procurement Lead**, I want a validated canned query for "top 5/10 OEMs over the last 18 months" so that the most repeated leadership question is available as a self-serve view.
- **D1-US2:** As a **Business Requester**, I want to ask "do we have a vendor for these goods/services?" and receive a role-appropriate spend-backed response so that I do not need to know procurement taxonomy.
- **D2-US1:** As a **Global Category Lead**, I want to ask "how much have we spent and committed with this vendor by category and region?" so that I can prepare evidence for negotiation and consolidation decisions.
- **D2-US2:** As a **Regional Procurement Lead**, I want drill-downs by month, supplier, category, region, cost centre, and PO/invoice source so that I can explain the answer rather than only quote a number.
- **D3-US1:** As a **Business Requester**, I want source attribution, freshness timestamp, and disclosure-appropriate detail on every spend answer so that I can trust the response without seeing commercially sensitive information I should not access.
- **D3-US2:** As a **Global Category Lead**, I want vendor overlap, committed spend, and cross-region spend concentration surfaced together so that I can identify consolidation opportunities.

Foundation 02 — Supplier Master + ESG

D1-F2.1 · Entity Resolution

- **D1-US1:** As an **AI Ops Analyst**, I want multi-signal matching (D-U-N-S, tax ID, address, fuzzy-name) with candidate pair scoring so that high-confidence matches auto-merge and edge cases come to me.
- **D2-US1:** As a **data scientist**, I want match-confidence logging and audit trail so that I can tune the matching model and investigate false merges.
- **D2-US2:** As a **PBP**, I want a single global supplier record post-resolution so that my Supplier 360 view reflects true consolidated spend.
- **D3-US1:** As a **Risk & Sustainability Lead**, I want resolved entities feeding sanctions screening so that screening isn't fragmented across duplicate records.

D1-F2.2 · Corporate Hierarchy Resolution

- **D1-US1:** As a **category strategist**, I want AI-built parent-child structures from corporate-tree data so that subsidiary spend rolls up to the global parent automatically.
- **D1-US2:** As an **AI Ops Analyst**, I want joint ventures and resellers flagged separately so that they don't distort parent-level aggregation.
- **D2-US1:** As a **PBP**, I want hierarchy-aware spend views so that negotiation leverage reflects total group spend, not just one subsidiary.
- **D3-US1:** As a **data scientist**, I want hierarchy confidence scoring so that ambiguous ownership structures are surfaced for human review.

D2-F2.3 · ESG & Risk Scoring

- **D2-US1:** As a **Risk & Sustainability Lead**, I want AI-driven ESG, diversity, financial-health, cyber-risk, and geographic-risk scores on the top 200 suppliers so that risk decisions are data-driven.
- **D2-US2:** As a **Risk & Sustainability Lead**, I want historical trend analysis so that I can see whether a supplier's risk posture is improving or degrading.
- **D3-US1:** As a **procurement lead**, I want red/amber/green tiering with automated escalation so that critical risk suppliers get immediate visibility.
- **D3-US2:** As a **PBP**, I want ESG scores visible in the Supplier 360 so that I bring risk context into every supplier meeting.

D3-F2.4 · Continuous Monitoring & Alerting

- **D3-US1:** As a **Risk & Sustainability Lead**, I want NLP-based ESG incident detection from news feeds so that I'm alerted to events before they hit formal ratings.
- **D3-US2:** As a **Risk & Sustainability Lead**, I want real-time sanctions screening so that compliance exposure is caught within hours.
- **D3-US3:** As a **procurement lead**, I want financial-distress signal detection so that supply continuity risks are flagged proactively.
- **D3-US4:** As an **AI Procurement Architect**, I want alert history and resolution tracking so that monitoring effectiveness can be measured over time.

Foundation 03 — Contract Intelligence (*persona anchor: Hooman asks for cross-region terms, commercials, expiry and co-term opportunities; Stacey asks whether a global contract exists and whether contracted pricing is being used; requesters ask whether a contract covers their need*)

D1-F3.1 · AI Contract Parsing

- **D1-US1:** As a **Global Category Lead**, I want active supplier contracts converted into structured metadata including parties, region, value, term, expiry, renewal terms, scope, and commercial terms so that I can compare coverage across regions.
- **D1-US2:** As a **Regional Procurement Lead**, I want global and regional contract coverage indexed by supplier and goods/services scope so that I can answer "is there a global contract for this vendor we can leverage?" quickly.
- **D2-US1:** As a **Global Category Lead**, I want terms and commercials compared across regions for the same vendor so that inconsistent positions and harmonisation opportunities are visible.
- **D2-US2:** As a **Regional Procurement Lead**, I want contracted price points linked to PO and invoice lines so that off-contract pricing can be detected before avoidable cost is committed.
- **D3-US1:** As an **Auditor / Contract Owner**, I want page-level traceability from every extracted field back to the source contract so that contract answers are defensible.

D1-F3.2 · Contract Validation (Human-in-the-Loop)

- **D1-US1:** As a **Commercial Value Architect**, I want a review interface for flagged ambiguities and low-confidence extractions so that I can correct and confirm efficiently.
- **D2-US1:** As a **Commercial Value Architect**, I want the ability to mark key clauses as canonical so that the system learns which clause variants are standard vs. non-standard.
- **D2-US2:** As a **data scientist**, I want every human correction fed back as training data so that the parsing model improves with each reviewed contract.
- **D3-US1:** As a **procurement lead**, I want validation progress tracking (reviewed vs. pending) so that I can monitor coverage of the top 200.

D2-F3.3 · Expiry, Co-term and Contract Coverage Monitoring

- **D2-US1:** As a **Global Category Lead**, I want contract expiry dates plotted across regions for the same vendor so that co-term negotiation windows can be identified.
- **D2-US2:** As a **Regional Procurement Lead**, I want contracts expiring within agreed action windows flagged so that tactical renewals do not silently roll over.
- **D3-US1:** As a **Global Category Lead**, I want spend, committed value, contract value, and expiry proximity compared so that leakage, underutilisation, and consolidation opportunities are visible.
- **D3-US2:** As a **Procurement Lead**, I want escalation rules for imminent expiry, missing owner, or unsupported contract coverage so that high-risk contracts receive senior visibility.

D2-F3.4 · Natural Language Contract Query

- **D2-US1:** As a **Commercial Value Architect**, I want to query the contract index in natural language ("auto-renewal contracts expiring in 6 months, value > \$1M") so that I get portfolio-level answers without building reports.
- **D2-US2:** As a **PBP**, I want contract status and key terms queryable alongside supplier and spend data so that I have full context in one place.
- **D3-US1:** As a **Front Door agent**, I want the contract index queryable via API so that "is there an MSA covering this?" resolves in seconds.
- **D3-US2:** As a **category strategist**, I want clause comparison queries across the portfolio so that I can identify non-standard terms at scale.

The Front Door — Clarification-led Multi-Agent Querying Layer (*architecture anchor: clarify before orchestration, execute through Spend, Supplier and Contract agents only, converge results in a join workspace, then synthesize with checker/repair and controlled fallback*)

D1-F4.1 · Request Intake, Structuring and Clarification Gate

- **D1-US1:** As a **Business Requester**, I want to ask procurement questions in plain English, such as "do we have a vendor for this?" or "is there a contract that covers this?", so that I can self-serve without understanding procurement taxonomy.
- **D1-US2:** As a **Regional Procurement Lead**, I want the Front Door to extract supplier, category, geography, time range, budget, and contract intent from a user question so that repeat tactical questions become structured requests.

- **D1-US3:** As a **System Architect**, I want a clarification gate before orchestration so that incomplete prompts are paused and targeted follow-up questions are asked before any specialist agent executes.
- **D2-US1:** As a **Business Requester**, I want the system to ask only the missing clarification, such as supplier, region, category, or time period, so that I do not have to restart the request.
- **D3-US1:** As a **Procurement Lead**, I want the original question, extracted structure, clarification history, and final structured request logged so that request interpretation can be audited.

D2-F4.2 · Orchestrator / Planner Agent

- **D2-US1:** As a **System Architect**, I want the orchestrator to select the right execution plan only after the clarification gate has produced a complete request so that agents do not guess missing intent.
- **D2-US2:** As a **System Architect**, I want the orchestrator to invoke only the relevant specialist agents — Spend, Supplier, and Contract — either in parallel or staged mode so that execution is controlled and source-specific complexity stays encapsulated.
- **D2-US3:** As a **System Architect**, I want the orchestrator to set and carry the aggregation grain, such as legal entity or parent supplier, so that downstream joins do not mix incompatible levels of detail.
- **D3-US1:** As a **Procurement Lead**, I want partial failures handled through a degrade path so that the answer can still be qualified when one source is stale, missing, or low quality.
- **D3-US2:** As an **Auditor**, I want the full execution trace — clarification, plan, agents called, source results, join logic, checker outcome, and fallback decisions — so that every answer is explainable.

D2-F4.3 · Spend Agent and Invoice Agent Access Pattern

- **D2-US1:** As an **Orchestrator**, I want the Spend/PO path to return committed spend, PO value, taxonomy, supplier canonical ID, region, category, and off-contract indicators so that committed exposure is available for cross-source synthesis.
- **D2-US2:** As an **Orchestrator**, I want the Invoice path to return posted spend, payment status, invoice timing, supplier canonical ID, region, and category so that actual spend can be compared with PO commitments and contract coverage.
- **D2-US3:** As a **Regional Procurement Lead**, I want the Spend Agent to support rolling-window and top-N supplier/OEM queries so that repeat leadership questions are answered without report stitching.
- **D3-US1:** As a **Global Category Lead**, I want spend and committed values returned on the same canonical supplier key as contracts and supplier master so that consolidation and co-term analysis are possible.
- **D3-US2:** As a **Data Scientist**, I want query latency, failed lookups, low-confidence classification, and fallback usage logged so that the spend access pattern can be tuned.

D2-F4.4 · Supplier Agent and Canonical Identity Access Pattern

- **D2-US1:** As an **Orchestrator**, I want the Supplier Agent to return canonical supplier ID, parent relationship, approved supplier status, hierarchy grain, risk flags, and ESG indicators so that cross-source joins are consistent.
- **D2-US2:** As a **Global Category Lead**, I want supplier hierarchy and vendor overlap surfaced so that subsidiaries, resellers, and duplicate supplier entities do not hide consolidation opportunities.
- **D2-US3:** As a **Regional Procurement Lead**, I want approved suppliers and framework status returned for a goods/services query so that I can answer "do we have a vendor for this?" quickly.
- **D3-US1:** As a **Business Requester**, I want supplier recommendations limited by my role and access level so that I see appropriate options without commercially sensitive detail.
- **D3-US2:** As a **Risk & Sustainability Lead**, I want amber/red supplier risk surfaced to the orchestrator so that high-risk suppliers are not silently auto-routed.

D2-F4.5 · Contract Agent and Structured Contract Lookup Pattern

- **D2-US1:** As an **Orchestrator**, I want the Contract Agent to return matching contracts, coverage scope, contract status, expiry date, renewal terms, supplier canonical ID, and source reference so that contract evidence can be joined with spend and supplier evidence.
- **D2-US2:** As a **Regional Procurement Lead**, I want the Contract Agent to answer "is there a global contract for this vendor?" with coverage limits and freshness cues so that I know whether it can be leveraged.
- **D2-US3:** As a **Global Category Lead**, I want cross-region terms and commercials returned in a comparable structure so that regional variation can be identified.
- **D3-US1:** As a **Business Requester**, I want to see whether an existing contract covers my need, with role-appropriate summary and next action guidance, so that duplicate sourcing is avoided.
- **D3-US2:** As a **Commercial Value Architect**, I want contract utilisation and ceiling indicators surfaced alongside contract match so that commitment and leakage risks are visible.

D2/D3-F4.6 · Join Workspace, Synthesis and Checker Loop

- **D2-US1:** As a **System Architect**, I want all specialist agents to emit rows keyed on canonical_supplier_id with an explicit grain flag so that PO, invoice, supplier, and contract evidence can be joined safely.

- **D2-US2:** As a **Global Category Lead**, I want the join workspace to combine committed spend, posted invoices, contract expiry, contract status, and supplier hierarchy so that questions such as "what have we committed to and where do contracts expire?" are answered from one evidence set.
- **D2-US3:** As a **Regional Procurement Lead**, I want the workspace to identify mismatches such as posted spend without active contract, contracted price versus PO variance, or supplier risk conflict so that intervention points are surfaced.
- **D3-US1:** As a **Business Requester**, I want the synthesized answer to include source, freshness, and role-appropriate limitation notes so that I understand what the answer is based on.
- **D3-US2:** As a **Procurement Lead**, I want the checker loop to separate missing-user-input issues from evidence-quality issues so that user gaps return to clarification while source-quality gaps return to orchestration, retry, re-scope, or qualified synthesis.
- **D3-US3:** As a **System Architect**, I want configurable loop limits for clarification and checker repair so that the agentic flow remains bounded and operationally predictable.

Expected Business Outcomes

Cost optimization across globe through multiple suppliers @Kalya n Chakravadhanula Check out the last bullet

- **Improve requester and procurement user experience:** Reduce friction in procurement intake and decision-making by making request status, route rationale, contract coverage, and next actions easier to understand for business requesters, procurement teams, and category stakeholders.
- **Increase operational speed and reliability:** Reduce cycle times, manual reconciliation effort, and avoidable delays across intake, sourcing, supplier evaluation, contract lookup, and route-to-action decisions.
- **Improve procurement decision quality:** Create a shared evidence base for prioritization, intervention, and trade-off decisions across spend visibility, supplier oversight, contract leverage, and guided request routing.
- **Strengthen governance, compliance, and trust:** Ensure that AI assistance is explainable, traceable, and aligned with procurement policy, risk, compliance, and human review requirements across sourcing and supplier decision flows.
- **Create a reusable foundation for broader procurement transformation:** Establish architecture, data, and agentic patterns that can scale from Horizon 1 foundations into broader procurement analytics, automation, and control-tower use cases.
- **Unlock measurable cost optimization:** Increase savings capture by surfacing supplier consolidation opportunities, committed-versus-historic spend, off-contract leakage, renewal timing, and benchmark price variance across the three personas. The expected result is stronger negotiation leverage for category leaders, faster tactical intervention by regional procurement, and better spend decisions at the point of requester intake.

Assumptions and restrictions to be considered

MVP (Wave 1) Assumptions and Scope Constraints

The MVP will focus on a clearly bounded set of procurement decision and orchestration areas during the initial 90-day delivery window. Scope is intentionally limited to spend intelligence, supplier resolution, contract intelligence, and Front Door routing. The MVP establishes the Procurement Command Center as a decision and intervention workspace for the highest-priority Horizon 1 capabilities.

Due to constraints on transforming the full procurement data estate within the MVP timeframe, the solution will initially operate on bounded integrations and tactical data grounding. Full enterprise-wide real-time integration, advanced optimization across all procurement journeys, and broader autonomous orchestration are deferred to later waves. AI capabilities in the MVP should focus on observability, explainability, triage, and next-best-action support, with all consequential decisions remaining human-led.

Key Considerations

- **Third party integrations:** All third party endpoints will be accessed via Azure API Management. The impact of MCP tools and applications, including latency trade offs, must be considered as part of the design.
- **Data responsibilities:** Any last mile data transformation required for the MVP should be implemented within the solution's resource groups, avoiding overlap or duplication with centralized enterprise data estate capabilities.
- **MVP data scope:** The MVP will focus on static data grounding only. Dynamic data feeds and real time integrations are explicitly deferred as a fast follow to later waves.
- **Infrastructure approach:** Infrastructure should follow a Just In Time, lean, and evolving architecture principle, provisioning only what is required for the MVP and expanding incrementally as capabilities mature.
- **Application layer:** Both functional and non functional requirements (usability, performance, security, and scalability) must be addressed in the application layer from the outset.
- **Innovation posture:** NTT Data leadership has set the direction to be at the forefront of innovation; therefore, selected preview features may be used where appropriate, provided risk and governance considerations are clearly managed.

Program level

The Procurement Command Center is designed to augment, not replace, human judgment, policy interpretation, and accountability. AI assistance supports observability, triage, prioritization, planning, and learning, while final ownership of sourcing, supplier, contract, and risk decisions remains with procurement and business stakeholders throughout all waves of the program.

Integrations with core ERP, procurement, sourcing, supplier, and contract systems are assumed to be delivered incrementally and will remain a key dependency for later waves. The Azure and data infrastructure should evolve from what is needed to support the MVP into what is required for the target-state architecture, rather than being fully provisioned upfront.

The program assumes a business-led design approach, with sustained engagement from procurement leaders, category stakeholders, supplier-risk owners, contract owners, and enabling functions. Active participation is required to validate priorities, review assumptions, shape agent behaviors, and ensure that the solution remains aligned to real operational needs rather than becoming a purely technical implementation.

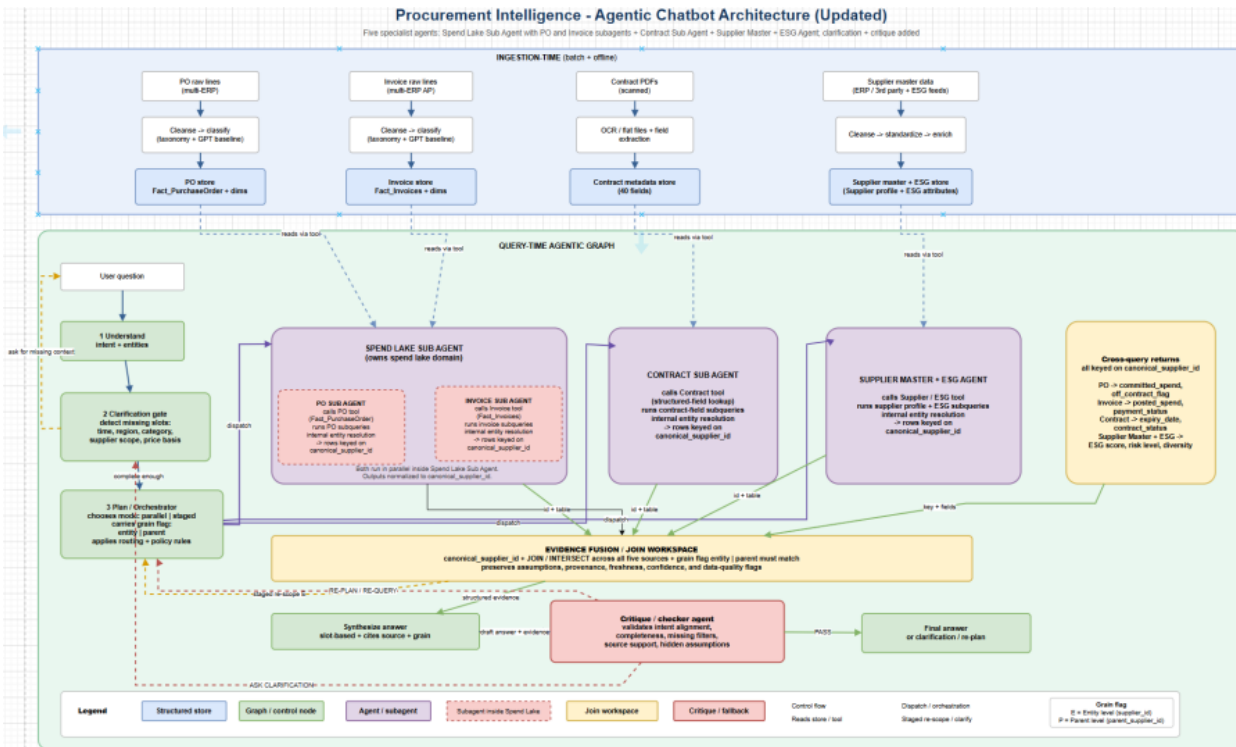
Definition of Done / Exit Criteria for MVP

The MVP will be considered complete when the following criteria are met:

- **Procurement Command Center is live and usable by business users:** The workspace supports procurement intake visibility, exception tracking, supplier and contract signals, and role-based intervention flows.
- **Defined MVP use cases delivered and validated with the business:** Horizon 1 procurement use cases are demonstrated and accepted by procurement and business stakeholders.
- **Data grounding implemented as agreed for MVP scope:** The agreed spend, supplier, contract, and intake signals are integrated to support procurement visibility, routing, and intervention.
- **Explainability and transparency validated:** AI-assisted summaries, triage outputs, and next-best actions clearly surface the underlying evidence and rationale.
- **Testing completed and quality baseline met:** Functional testing, basic performance validation, and user acceptance testing are completed for all MVP components.
- **End-user feedback collected and incorporated:** Feedback from procurement users, category stakeholders, and related business users is gathered during MVP usage and documented to inform later waves.
- **Operational handover completed:** NTT DATA can operate, support, and maintain the MVP environment, including monitoring, incident handling, and knowledge transfer.

Proposed Solution

Conceptual Solution Design



The solution separates what is precomputed offline from what happens live when a user asks a question, while adding an explicit clarification gate before orchestration. All heavy data engineering — cleansing, classification, OCR, extraction, and canonical key preparation — remains in batch. At query time, the agent layer first structures the user request, asks for missing information when needed, and only then reads prepared stores through specialist agents. This keeps live latency low, prevents incomplete prompts from reaching execution, and makes correctness reproducible because agents never reshape raw data on the hot path.

Offline foundation: Three independent pipelines

Each source is ingested through its own three-stage path — raw → cleanse/classify → structured store — and the three never touch each other during ingestion. That isolation means any source can be re-ingested, re-classified, or fail without blocking the others.

- **PO:** multi-ERP purchase-order lines → cleanse + classify (taxonomy primary, GPT as baseline/fallback classifier) → Fact_PurchaseOrder + dimensions.
- **Invoice:** multi-ERP AP lines → the same cleanse/classify logic → Fact_Invoices + dimensions. PO and Invoice deliberately share one dimensional model, so identical classification applies to both.
- **Contract:** scanned PDFs → OCR + field extraction → contract metadata store (40 fields).

The structural asymmetry is the key risk signal: PO and Invoice land as structured facts on a shared model, but Contracts must be turned from pixels into fields. The Contract branch carries fundamentally more extraction risk than the other two — it is the only path converting unstructured documents into queryable structure.

Live request flow with clarification gate

Front end and orchestration path:

1. User question comes in through the procurement front door.
2. Understand / structure request — parse intent, extract entities, identify category, geography, supplier, time range, budget, and contract intent.
3. Clarification gate — check whether the structured request is complete enough to execute. If required fields are missing or ambiguous, the system asks a targeted clarification question and pauses execution until the missing input is supplied.
4. Plan / Orchestrator — once the request is complete, the orchestrator chooses execution mode, selects the relevant specialist agents, fixes the aggregation grain (entity or parent), and carries that grain flag through the rest of the run.

Specialist subagents remain the encapsulation layer. They are invoked only after the clarification gate has produced a complete, execution-ready request, and each subagent follows the same contract:

- calls its own tool (PO over Fact_PurchaseOrder, Invoice over Fact_Invoices, Contract over a structured-field lookup),
- runs its own subqueries,
- does its own internal entity resolution,
- and returns rows keyed on canonical_id.

Each subagent is a self-contained specialist that hides its source's messiness and exposes one clean thing: rows on a common key. The orchestrator does not reach into a store directly; the only way into a source is through that source's subagent tool. A degrade path wraps the subagents to handle missing, stale, or low-quality data, so a weak source produces a flagged-but-graceful answer rather than a hard failure.

Convergence — the join workspace: All three subagents emit their keyed rows into a single neutral join workspace, which performs JOIN / INTERSECT across all three on canonical_supplier_id. Two correctness guards live here:

- the cross-source join happens in the workspace, not inside any one tool — so no single source dictates the combined view;
- the grain flag must match — entity-grain rows can't be intersected with parent-grain rows, so the grain chosen up front is enforced at the join.

Each source contributes a defined slice to the combined row: PO → committed spend off contract flag; Invoice → posted spend + payment status; Contract → expiry date contract_status. This is what lets the solution answer genuinely cross-source questions ("which suppliers have posted invoices but no active contract") that no single store could answer alone.

Synthesis and checker loop: The joined result feeds a slot-based answer that cites source and grain for every value. A critique/checker step then validates whether the answer is grounded, internally consistent, and sufficiently complete. If the issue is missing user intent, the flow returns to the clarification gate. If the issue is source quality, stale data, failed retrieval, or inconsistent agent outputs, the flow returns to the orchestrator to re-plan, retry, narrow scope, or produce a qualified answer with explicit limitations.

What makes it agentic: clarification, staged execution, and controlled fallback

- **Clarification loop:** incomplete or ambiguous prompts are handled before execution. The system asks a targeted question for missing supplier, category, geography, time range, budget, or contract intent instead of allowing the orchestrator to guess.
- **Staged re-scope:** after a first pass yields a key set E, the workspace can hand that set back to the orchestrator to narrow the next round of subagent queries to just those suppliers. This is the staged mode — cheaper, sequential, and dependency-aware — versus the parallel mode where relevant subagents fire at once.
- **Degrade / fallback:** the wrapper around the subagents lets the system answer with partial coverage and explicit quality flags when one source is stale, missing, or low confidence.
- **Critique and repair loop:** the checker does not duplicate the clarification gate. It separates user-input gaps from execution or evidence gaps, sending the former back to clarification and the latter back to orchestration or qualified synthesis.

2.2. Data Requirements

The effectiveness of the procurement intelligence workstream depends on building a reliable procurement data layer across **spend, suppliers, contracts, and incoming demand**. For Horizon 1, this is less about creating an all-new enterprise data estate and more about establishing decision-grade data products that can power routing, query, and monitoring use cases.

For the MVP, data requirements should focus on the three Horizon 1 foundations and the Front Door, with later expansion into richer sensing, optimization, and broader automation scenarios.

2.2.1. Data Sources - MVP

Spend and transaction systems

- SAP and other ERP transaction records
 - purchase orders
 - invoice lines
 - supplier references, amounts, currencies, cost centers, and GL attributes
- Ariba and sourcing platform data
 - sourcing events
 - catalog and procurement context
 - request and supplier touchpoints where available

Supplier master and enrichment sources

- supplier master records from ERP and procurement systems
- parent-child and legal entity relationship data
- ESG, diversity, sanctions, financial health, cyber, and geographic risk signals for priority suppliers

Contract sources

- contract repositories and shared drives
- active supplier agreements, amendments, and framework documents
- metadata needed for extraction, indexing, and expiry monitoring

Front Door and workflow inputs

- incoming procurement requests from forms, email, collaboration tools, or intake channels
- structured request fields such as category, geography, urgency, budget, and contract intent
- routing outcomes, overrides, and feedback signals for evaluation

Monitoring and evaluation data

- audit samples and gold sets for spend classification, supplier resolution, contract extraction, and routing evaluation
- confidence, override, regression, and drift metrics
- latency and model health signals for operational readiness

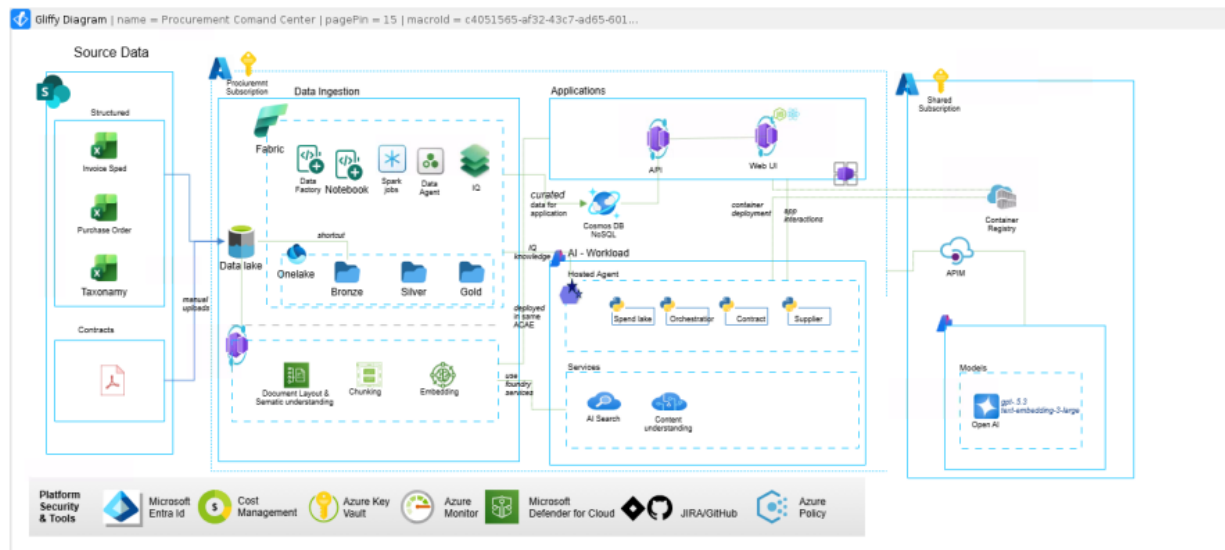
Wave 2+ data expansion

- broader demand-sensing inputs
- expanded supplier and contract coverage beyond the top 200
- deeper analytics for anomalies, off-contract spend, and optimization opportunities

Key Data Principles

- Start with bounded, decision-grade Horizon 1 foundations rather than full enterprise unification
- Preserve cross-ERP traceability across spend, supplier, contract, and request context
- Ensure outputs are confidence-scored, explainable, and auditable
- Design for human review where uncertainty, risk, or incomplete data is material

Architecture



2.3.1. Layered Capability Architecture

The architecture should follow a layered procurement intelligence model, aligned to the conceptual approach. The live path starts with the Front Door and clarification gate, then hands complete requests to an orchestrator that consults three specialist foundation agents: Spend Lake Agent, Supplier Agent, and Contract Agent. The design should not assume a hosted-agent construct for the specialist layer; the agents are logical capability blocks that expose governed access to the underlying spend, supplier, and contract foundations.

Layer 1: Command Center (Observability and Decision Layer)

- Single workspace for:
 - procurement intake visibility

- request and exception monitoring
- supplier and contract signal tracking
- decision support and exception management
- Role-based views:
 - Category Strategist
 - Procurement Business Partner
 - Procurement Operations
 - Business Stakeholders

Layer 2: Agentic Orchestration and Foundation Agent Layer

- Agents embedded into the request and evidence flow:
 - Front Door intake and clarification agent
 - Orchestrator / routing planner
 - Spend Lake Agent
 - Supplier Agent
 - Contract Agent
- Capabilities:
 - request structuring and targeted clarification before execution
 - routing and plan generation across the three foundation agents
 - grounded retrieval from spend, supplier, and contract foundations
 - cross-source synthesis, critique, and controlled fallback when data is incomplete or inconsistent

Layer 3: Data and Integration Fabric

- Hub-and-spoke integration and governed data access across:
 - Spend Lake — purchase orders, invoices, taxonomy classification, and spend facts
 - Supplier foundation — supplier master, entity resolution, hierarchy, ESG, and risk signals
 - Contract foundation — contract repositories, extracted metadata, clause index, and expiry signals
 - Front Door workflow and request-state data

Responsibilities:

- batch and incremental ingestion into the three foundations
- API and tool access for Spend Lake Agent, Supplier Agent, and Contract Agent
- source freshness, lineage, and dependency tracking
- join-ready canonical supplier identifiers and grain controls for cross-source synthesis

Layer 4: Native Execution Systems

Systems continue to execute business logic:

- ERP systems - transaction execution and financial posting
- sourcing systems - sourcing event and catalog processes
- contract systems - contract storage and lifecycle processing
- supplier and risk platforms - supplier and compliance data updates

The command center does not replace systems - it orchestrates and observes them.

Layer 5: Employee Experience Layer (Single Front Door)

- unified experience for:
 - business requesters
 - procurement users
 - stakeholders and approvers

Capabilities:

- intake and request workspace
- procurement command center workspace
- guided decision and routing support
- conversational procurement assistant interface

Key Architectural Principles

- Clarify before orchestrating; do not execute incomplete prompts.

- Use three foundation agents only: Spend Lake Agent, Supplier Agent, and Contract Agent.
- Keep agents as governed access layers over prepared foundations, not replacements for source systems.
- Observe -> Orchestrate -> Assist, with human-in-the-loop decisions for material risk or uncertainty.
- Incremental integration from MVP to scale.

2.4. Service/component mapping to function blocks

Interface

- Container Apps

Agentic Crew

- Agents
 1. Container Apps
 2. Foundry - AI models
- Toolkits Hub
 1. Foundry
 2. Foundry IQ
 3. AI Search
 4. Langgraph

Data Integration

- Orchestration
 - Microsoft Fabric - Pipelines
 - Schedules - Pipeline scheduling
 - Metadata - SQL DB or Delta on Lakehouse on Fabric
- Data Engineering
 - Microsoft Fabric - Spark Jobs built using Python
- Persistence
 - Microsoft Fabric - Lakehouse - Medallion
 - Fabric IQ - Ontology, Knowledge Graph
 - Azure Storage Account
 - Data Lake Storage Gen 2 - Data Landing
 - Blob - Application Logs
 - Azure Cosmos DB - Application/Agent metadata, state management
- Analytics
- ML

Observability

- Foundry

Identity & Security

- Microsoft Entra ID
- Azure Key vault

Application telemetry

- Application Insights

Data governance

- Microsoft Purview

DevOps

- ~~Azure DevOps~~
- JIRA
- Confluence
- Github Repo
- Github Actions

- Automated testing in CI/CD pipeline
- Github Copilot

Implementation

Data Preprocessing

The preprocessing layer is responsible for transforming fragmented procurement data into a **consistent, interpretable, and actionable foundation** for downstream analytics, routing, and agentic workflows.

3.1.1. Input Data and Assumptions

The MVP will operate on:

- cross-ERP spend transaction data
- supplier master and hierarchy data
- contract documents and extracted fields
- sample and live procurement requests for Front Door routing
- evaluation and feedback signals from analysts and business users

Constraint: Data originates from multiple systems with inconsistent structures, identifiers, coverage, and freshness. Preprocessing must normalize and align these signals without requiring full system transformation during Horizon 1.

3.1.2. Data Preprocessing Objectives

1. Spend normalization and taxonomy alignment

- Normalize currencies, entities, supplier references, and transaction fields across source systems
- Map transactions to the NTT DATA taxonomy with confidence scoring
- Create a unified spend surface for analytics and downstream routing

2. Supplier entity resolution

- Standardize supplier identifiers, names, and legal entity attributes across systems
- Resolve duplicates and parent-child relationships
- Attach enrichment and risk attributes to priority suppliers

3. Contract extraction and indexing

- Extract key fields such as parties, term, value, expiry, and critical clauses
- Create queryable indexes and monitor upcoming expiries
- Link contracts to suppliers and spend context

4. Request structuring and routing preparation

- Extract request intent, category, geography, budget, and urgency from intake signals
- Prepare inputs for parallel consultation of spend, supplier, and contract foundations
- Detect ambiguity, missing fields, stale data, and low-confidence routing conditions

5. Evaluation, exception detection, and human review

- Identify low-confidence classifications, unresolved supplier matches, parser ambiguities, and routing exceptions
- Route these cases for analyst or business review
- Use corrected outcomes to improve the models and thresholds

6. Procurement decision-support signal construction

- Combine:
 - confidence scores from spend classification, supplier resolution, contract extraction, and routing
 - data quality, freshness, and coverage indicators across the three foundations
 - human overrides, review outcomes, and exception patterns from the Front Door workflow

- Generate:
 - decision-support confidence and escalation signals
 - explanations that show why a request can be auto-routed, requires review, or should be flagged for risk or data-quality concerns

3.1.3. Data Outputs for Command Center

- - - unified procurement request and evidence view
 - classification, supplier, contract, and routing status signals
 - cross-source dependency and provenance view
 - exception flags for low confidence, missing data, and policy or risk concerns
 - decision and review readiness signals for procurement users

3.1.4. Role of Data Science

- - - define classification, routing, and escalation logic
 - build risk and anomaly detection models across spend, supplier, and contract signals
 - create prioritization signals for procurement review queues and exception triage
 - building agents
 - enable **explainable agent outputs**

MLOps & DataOps

Responsible AI

The Customer acknowledges that this engagement includes the development or deployment of an AI system that: (1) is powered by AI and Customer is being exposed to and interacting with an AI system, (2) the AI system output may contain sensitive content and/or factual inaccuracies, (3) Customer is responsible for determining the accuracy of any content generated by this AI system in relation to Customer's intended use as the user of this AI system, and (4) this AI system does not provide opinions or advice. It is not designed to replace the role of qualified procurement, legal, finance, risk, or human reviewers. Customer is solely responsible for displaying and/or obtaining appropriate consents, warnings, disclaimers, and acknowledgements to end users of Customer's implementation of the AI system. During the engagement, if the requested business outcome includes Microsoft developing or deploying an AI System for or with Customer which may be considered a sensitive use, Microsoft will conduct an internal responsible AI review, to include assessment of and requirements for the potential sensitive use.

The outcome of Microsoft's review will be comprehensively discussed with NTT Data, ensuring a transparent understanding of the findings. Microsoft is dedicated to adhering to its RAI principles, and as such, will identify and implement any necessary changes. For more detailed information about Microsoft's commitment to responsible AI, please refer to <https://aka.ms/RAI>.

Should the review uncover any discrepancies between the use-case and the RAI guidelines, Microsoft will engage with NTT DATA to co-develop a robust mitigation plan in addressing and resolving potential risks, ensuring that the project is brought into alignment with the RAI standards. The process may lead to modifications of the current project or proposal of an alternative solution that complies with the RAI expectations.

Project Team, Roles, Governance

Roles Required

Role	Definition
Architect Product Manager	This role acts as a business-focused product manager, responsible for translating procurement goals, Horizon 1 scope, workflow priorities, and user needs into end-to-end solution design, backlog, and measurable outcomes. This goes beyond delivery process management and focuses on building the right procurement foundation and workflow proof incrementally with continuous stakeholder feedback.
UX designer	Designs enterprise-grade, decision-centric UX for procurement users, balancing AI assistance, explainability, and user control across spend, supplier, contract, and Front Door experiences. Must have experience with enterprise role-based UX.
UI engineer	App Consultant can convert static UX mockups into basic UI. But this procurement workstream is a decision platform with complex human-in-the-loop workflows. Without an experienced UI engineer, the app layer may become a bottleneck and may push workflow complexity into the backend, which would not be an optimal design choice.
Data scientists	Responsible for designing and implementing the decision-intelligence logic of the procurement workstream, including spend classification, supplier resolution, contract extraction evaluation, routing logic, confidence calibration, anomaly detection, and explainability of AI-assisted recommendations.

Data engineer	Responsible for designing and validating the data integration and canonical data layer using Microsoft Fabric, requiring hands on experience with Fabric and proven enterprise delivery to ensure reliable, governed, business ready data foundations for the platform.
App consultants (Backend engineers)	Strong Python/FastAPI backend engineering with Azure OpenAI integration, Agent Framework-style agent orchestration (+MCP), and production-grade async workflow, deployment, and observability practices.
Infra (IaC, CI/CD)	Responsible for provisioning and operating the platform runtime across Azure and Microsoft Fabric, covering Infrastructure as Code and CI/CD for both application and data layers. The role requires experience with Terraform, Azure DevOps or GitHub Actions, Azure Container Apps / App Service, Application Insights, and Microsoft Fabric (workspace and capacity provisioning, CI/CD for Fabric artifacts), as well as identity and access management via Entra ID and RBAC.
Security consultant	Responsible for defining and enforcing security, governance, and Responsible AI controls across the platform, including identity and RBAC, auditability, data access policies, and human approval gates
Testers	Responsible for validating end to end decision flows in the application layer and requires a solid understanding of the marketing business scenarios to assess whether the platform supports clear, correct, and usable decisions, not just technical correctness.
Project manager	Responsible for overall delivery coordination, dependencies, and stakeholder alignment.
Scrum master	Focuses on enabling effective team execution through agile ceremonies, flow, and continuous improvement, working closely with the customer Product Owner to support delivery alignment and feedback loops, while not owning product scope, solution design, or business decision trade offs.

4.2. Definition of Ready for MVP

Requirement	Status
Business requirements are complete and approved	
Data requirements are finalized which means all required datasets for MVP use cases are identified (Data Sources.xlsx)	
Technical preconditions are met: ○ Required Azure environments for MVP development and testing are provisioned ○ All necessary access permissions granted to delivery teams (NTT Data env access status.xlsx) ○ Technical constraints and architecture are documented etc.	
User and stakeholder alignment completed	
Change requirements identified (if any)	
Operational readiness	
Risks, dependencies and assumptions are identified and logged	

4.3. Sprints Execution: High Level Plan

This section explains how delivery will run rather than repeating the capability roadmap. The recommended operating model is a two-week sprint cadence with Sprint 0 used to establish feasibility evidence, refine the MVP backlog, confirm data readiness, and validate the thin-slice design assumptions before committed delivery begins. From D1 onward, each sprint should follow a consistent rhythm: backlog refinement with procurement stakeholders, design and build of a limited set of thin-slice features, data and model validation, end-to-end demonstration, and explicit acceptance of what moves forward versus what remains in review.

- **Sprint cadence:** Two-week sprints, with Sprint 0 used as the feasibility and readiness gate before D1, D2, and D3 execution.
- **Planning and refinement:** Weekly backlog refinement with the architect product manager, delivery team, and business stakeholders to keep the sprint backlog aligned to Horizon 1 priorities.
- **Build and validate loop:** Each sprint should include data preparation, feature implementation, model or rules validation, UX review where relevant, and an end-to-end demo on real procurement scenarios.
- **Governance rhythm:** Weekly delivery checkpoint, sprint review with business stakeholders, and a decision forum for scope trade-offs, risk escalation, and acceptance of thin-slice outputs.
- **Acceptance approach:** Features should only progress when the underlying data, routing logic, and explainability are sufficient for business review; unresolved quality or policy issues remain in controlled review queues rather than being hidden in the roadmap.

4.4. High Level Schedule/Timeline

Sprint 0 / F	Weeks 0-2	Feasibility evidence, backlog shaping, data and access readiness, thin-slice validation	Go / no-go into D1
D1	Weeks 3-6	Production thin slice for core foundations and initial Front Door flows	Usable thin-slice demo and quality baseline
D2	Weeks 7-10	Scale-out, convergence of foundations, stronger controls and routing logic	Business review of scaled capability set
D3	Weeks 11-14	Operational hardening, monitoring, explainability, and readiness for handover	MVP exit review and Wave 2 recommendations

4.5. MVP: 90 Days Approach

The 90-day MVP should be understood as a staged proof of value rather than a full procurement transformation. By **Day 30**, the program should have exited Sprint 0 and established a working thin slice across the most critical data and routing assumptions, with priority backlog, access, and success measures agreed. By **Day 60**, the focus should be on extending the thin slice into a usable cross-foundation experience, improving model and routing quality, and validating the priority personas and use cases with business stakeholders. By **Day 90**, the objective is to demonstrate an operational MVP with usable monitoring, explainability, and clear business value signals across spend, supplier, contract, and Front Door workflows, while explicitly identifying what remains for Wave 2.

- **Day 30:** feasibility evidence complete, scope and backlog confirmed, core thin-slice path demonstrated.
- **Day 60:** foundations converging, routing quality improving, and business users validating the first usable experience.
- **Day 90:** MVP ready for exit review with measurable outcomes, monitoring, explainability, and a defined Wave 2 backlog.

4.6. Velocity Impact Factors

- **Environment and access readiness:** Delays in tenant access, security approvals, tool provisioning, or data connectivity can materially slow Sprint 0 and push downstream delivery.
- **Data quality and coverage:** Incomplete taxonomy mapping, fragmented supplier identifiers, low-quality contract scans, or inconsistent metadata can increase review effort and reduce delivery velocity.
- **Business stakeholder availability:** The speed of backlog clarification, design decisions, and acceptance reviews depends on timely input from procurement, category, contract, and risk stakeholders.
- **Cross-team dependencies:** UX, engineering, data, security, and integration workstreams need to stay aligned; unresolved dependency bottlenecks will reduce effective sprint throughput.
- **Human-in-the-loop design effort:** Review queues, confidence thresholds, exception paths, and explainability controls are essential for quality but may slow feature completion if not designed early.
- **Integration and policy constraints:** API limitations, security requirements, data governance controls, and approval gates may affect how quickly features can move from prototype to accepted MVP capability.

Risks, Issues and Dependencies

S. No	Title	Owner	Risk/issue Management Plan		Due Date
1.	Environment and access readiness		Confirm all required environments, identities, RBAC, and data-access paths before Sprint 0 exit; track blockers weekly and escalate unresolved access dependencies.		Sprint 0 exit
3.	Data quality and taxonomy consistency		Prioritize a bounded thin slice, define taxonomy mapping rules early, and create review queues for low-confidence classification and unresolved supplier/entity mappings.	D1 baseline	
4.	Contract document quality and extraction variability		Sample contract populations early, separate clean digital files from scanned or poor-quality documents, and route low-confidence extractions into human validation before relying on them operationally.	D1-D2	
5.	Supplier master fragmentation and hierarchy resolution		Start with top suppliers and priority categories, define matching signals and confidence thresholds, and make ambiguous parent-child relationships visible for business review rather than forcing automated rollups.	D1-D2	
6.	Source-system integration and API dependency timing		Use bounded integrations and static or staged data grounding where necessary for MVP; document deferred dynamic feeds and avoid blocking Horizon 1 on full enterprise connectivity.	Sprint 0 scoping / D1	
7.	Human-in-the-loop operating model not defined early enough		Define review queues, confidence thresholds, exception paths, and role responsibilities during Sprint 0 so delivery does not optimize only the AI path while leaving operations unresolved.	Sprint 0 / D1	
8.	UX and application-layer dependency for Front Door workflows		Prioritize the minimum viable intake, triage, and review experience early; avoid pushing unresolved workflow complexity into backend logic due to delayed UX or UI decisions.	D1	
9.	Scope expansion beyond Horizon 1 MVP		Keep Horizon 1 bounded to the agreed foundations and Front Door proof workflow; route additional asks such as broader orchestration, full dynamic integrations, and deeper ERP extensions into a controlled Wave 2 backlog.	All phases	